1 Site Displacement and Geophysical Corrections

The receiver coordinate that enters the line-of-sight construction and the linearised geometric range of the Foundations document is not the instantaneous crust position: it is the conventional, tide-free station position to which the IGS/ITRF products are referenced, displaced by a sum of *a priori* geophysical corrections. Each correction is a deterministic, modelled 3D displacement $\boldsymbol{d}(t)$ of the antenna reference point. None is estimated; all are computed from the (cm-grade) a priori position, the ephemerides of the Sun and Moon, and Earth-orientation parameters, then projected onto the line of sight and folded into the modelled range. This document derives those models as the *intended* IERS 2010 convention [2], so that it serves as the oracle that resolves the sign and phase questions the implementation has surfaced.

Write the corrected receiver position as

$$\boldsymbol{r} = \boldsymbol{r}_0 + \boldsymbol{d}_{\mathrm{SET}} + \boldsymbol{d}_{\mathrm{OTL}} + \boldsymbol{d}_{\mathrm{PT}} + \boldsymbol{d}_{\mathrm{OPT}} + \boldsymbol{d}_{\mathrm{APL}}, \tag{1}$$

with $\boldsymbol{r}_0$ the conventional a priori position and each $\boldsymbol{d}$ a millimetre-to-decimetre ECEF vector. The five terms are the solid Earth tide ($\boldsymbol{d}_{\mathrm{SET}}$, the largest, up to $\sim$30 cm radial), ocean tidal loading ($\boldsymbol{d}_{\mathrm{OTL}}$, coastal cm), the rotational-deformation (solid) pole tide ($\boldsymbol{d}_{\mathrm{PT}}$, sub-cm to cm secular-plus-Chandler), the ocean pole tide ($\boldsymbol{d}_{\mathrm{OPT}}$, sub-mm to $\sim$2 mm), and atmospheric pressure loading ($\boldsymbol{d}_{\mathrm{APL}}$, 2–20 mm). The last is presently a known gap (Section 1.6). All five are computed in `src/core/corrections/GnssCorrections.cpp` and `src/core/corrections/OceanTideModel.cpp`; the LOS projection and summation into the modelled range live in `src/core/model/ModelGeometry.cpp` and `src/core/model/PppModel.cpp`. Each term carries a config flag (`enable_solid_earth_tide`, `enable_ocean_tide_loading`, `enable_pole_tide`, `enable_ocean_pole_tide`); with a flag off, that term contributes a zero vector and the result reduces to the uncorrected position.

1.1 Sign convention and projection into the range

Every displacement is projected onto the unit line of sight before it enters the modelled range. Let $\boldsymbol{e} = (\boldsymbol{r}^i - \boldsymbol{r})/\|\boldsymbol{r}^i - \boldsymbol{r}\|$ be the receiver-to-satellite unit vector (the convention of the Foundations LOS equation; see `ModelGeometry.cpp` where `los = (sat_pos -rx_pos)/range`). A site displacement $\boldsymbol{d}$ changes the geometric range by the component of $\boldsymbol{d}$ *along* the receiver-to-satellite direction with a sign that shortens the range when the antenna moves *toward* the satellite. Concretely the scalar correction summed into the modelled range is

$$\boxed{\delta\rho \;=\; \boldsymbol{d}\cdot(-\boldsymbol{e}) \;=\; \boldsymbol{d}\cdot\frac{\boldsymbol{r}-\boldsymbol{r}^i}{\|\boldsymbol{r}-\boldsymbol{r}^i\|},} \tag{2}$$

so a displacement that pushes the receiver *toward* the satellite (a positive component along $\boldsymbol{e}$) yields $\delta\rho < 0$ and shortens the modelled range, as required. The ocean-loading and pole-tide projections use exactly this form (`ModelGeometry.cpp`: `atm_loading_corr = otl_ecef.dot(-los)`; `PppModel.cpp`: `pole_tide_ecef.dot(-los)`), and the modelled range adds them in (`PppModel.cpp`, the `computeBaseModeledValue` return sums `+ atm_loading_corr + solid_tide` and the pair processors add `+ pole_tide_corr`). The solid-Earth-tide helper performs its own projection internally with the algebraically identical vector $(\boldsymbol{r}-\boldsymbol{r}^i)/\|\boldsymbol{r}-\boldsymbol{r}^i\|$ (`GnssCorrections.cpp`, `computeSolidEarthTide`: `los = (receiver_pos -sat_pos)/rho_norm`, then `disp.dot(los)`), so all four tide terms share

one consistent sign. Throughout this document we therefore give each model as a 3D ECEF (or local) displacement $\boldsymbol{d}$ and leave Equation (2) implicit.

The Sun and Moon positions $\boldsymbol{r}^{\odot}, \boldsymbol{r}^{\mathrm{m}}$ on which the body tides depend, and the ECI↔ECEF rotation and Earth-orientation parameters they require, are produced by `src/core/model/Planetary.cpp` and are documented in the forthcoming *Coordinate Frames and Time Systems* reference; here we take $\boldsymbol{r}^{\odot}, \boldsymbol{r}^{\mathrm{m}}$ as given geocentric ECEF vectors.

1.2 Solid Earth tide (IERS 2010, Steps 1 + 2)

The solid Earth tide is the elastic deformation of the whole Earth under the lunisolar tide-generating potential. The implementation is a faithful port of the IERS Conventions `dehanttideinel` routine [2], evaluated in two steps: a time-domain Step 1 from the instantaneous Sun/Moon geometry, plus a frequency-domain Step 2 that corrects the small handful of constituents whose Love/Shida numbers depart from the nominal degree-2 values. **This entire Step 1+2 block is oracle-validated**: it reproduces the official IERS `dehanttideinel` test vector and the in-tree POINT reference to sub-millimetre, cross-checked by `scripts/solid_tide_reference.py` (`GnssCorrections.cpp:1196--1201`). The permanent (zero-frequency) tide is *retained*, so the corrected coordinate is conventional tide-free, matching IGS/ITRF (`GnssCorrections.cpp:1200--1201`).

1.2.1 Step 1 — in-phase degree 2 and 3

For each tide-raising body $b \in \{\odot, \mathrm{m}\}$ at geocentric position $\boldsymbol{r}^b$ ($r_b = \|\boldsymbol{r}^b\|$) and the station at $\boldsymbol{r}_0$ ($r = \|\boldsymbol{r}_0\|$, geocentric latitude φ via direction cosines), the degree-2 in-phase displacement is built from the cosine of the body zenith angle $\cos Z_b = \hat{\boldsymbol{r}}_0 \cdot \hat{\boldsymbol{r}}^b$ and a body scale factor

$$\mathrm{fac}_b \;=\; \frac{M_b}{M_\oplus}\, R_e \left(\frac{R_e}{r_b}\right)^3, \tag{3}$$

with $R_e = 6378136.55$ m and mass ratios $M_\odot/M_\oplus = 332945.943062$, $M_\mathrm{m}/M_\oplus = 0.012300034$ (`GnssCorrections.cpp:1211--1213, 1441`). The radial and tangential weights are

$$p_2^b = 3\Big(\tfrac{h_2}{2} - l_2\Big)\cos^2 Z_b - \tfrac{h_2}{2}, \tag{4}$$

$$x_2^b = 3\, l_2 \cos Z_b, \tag{5}$$

(`GnssCorrections.cpp:1431, 1435`), where h_2, l_2 are the *latitude-dependent* degree-2 Love and Shida numbers of IERS Eq. 7.2,

$$h_2 = 0.6078 - 0.0006\,(1 - \tfrac{3}{2}\cos^2\varphi), \qquad l_2 = 0.0847 + 0.0002\,(1 - \tfrac{3}{2}\cos^2\varphi) \tag{6}$$

(`GnssCorrections.cpp:1424--1425`). An analogous degree-3 term uses fixed $h_3 = 0.292$, $l_3 = 0.015$ and $\mathrm{fac}_b^{(3)} = \mathrm{fac}_b\,(R_e/r_b)$ (`GnssCorrections.cpp:1210, 1433--1438, 1442`). The Step-1 in-phase displacement sums the radial weight along the station unit vector and the tangential weight along each body's unit vector:

$$\boxed{\boldsymbol{d}^{(1)}_{\mathrm{SET}} = \sum_{b\in\{\odot,\mathrm{m}\}} \mathrm{fac}_b\Big(x_2^b\,\hat{\boldsymbol{r}}^b + p_2^b\,\hat{\boldsymbol{r}}_0\Big) \;+\; (\text{degree-3 analogue}),} \tag{7}$$

matching the per-component accumulation in `GnssCorrections.cpp:1448--1453`.

1.2.2 Step 1 — out-of-phase anelasticity and the L^1 term

Mantle anelasticity makes the diurnal and semidiurnal degree-2 Love numbers complex; the small imaginary parts produce out-of-phase displacements `ST1IDIU` (diurnal, $\delta h = -0.0025$, $\delta l = -0.0007$) and `ST1ISEM` (semidiurnal, $\delta h = -0.0022$, $\delta l = -0.0007$) (`GnssCorrections.cpp:1219, 1240`). The latitude (L^1) dependence of the Love numbers contributes a horizontal-only correction `ST1L1` with $l^{(1)}_{\rm diu} = 0.0012$, $l^{(1)}_{\rm sem} = 0.0024$ (`GnssCorrections.cpp:1265`). These three helpers (`GnssCorrections.cpp:1216--1302`) return local radial/north/east displacements rotated into ECEF and are added to Equation (7) at `GnssCorrections.cpp:1456--1458`; their detailed trigonometric forms follow the IERS reference verbatim and are not reproduced here.

1.2.3 Step 2 — frequency dependence (Doodson tables)

Step 2 corrects the constituents whose actual Love numbers differ from the nominal degree-2 value, summing harmonic terms keyed by the Doodson fundamental arguments. The astronomical arguments τ, s, h, p, N', p_s (lunar time, Moon/Sun mean longitudes, lunar perigee, negative node, solar perigee) are polynomials in Julian centuries T from J2000, with the 15°/hr Earth-rotation term entering τ through the fractional UT hour H:

$$s = 218.31664563^\circ + 481267.88194\,T - 0.0014663889\,T^2 + \ldots, \tag{8}$$

$$\tau = 15^\circ H + 280.4606184^\circ + 36000.7700536\,T + \ldots - s, \tag{9}$$

with the remaining h, p, N', p_s polynomials given to fourth order in `GnssCorrections.cpp:1306--1324` (`computeAngles`). For each table line the phase is $\theta = \tau + n_s s + n_h h + n_p p + n_{N'} N' + n_{p_s} p_s (+\,\lambda$ for diurnal) and the in-/out-of-phase radial and tangential amplitudes (millimetres) are applied with a latitude weight.

Diurnal band (Table 7.3a). Thirty-one lines, dominated by K_1 and P_1 (`GnssCorrections.cpp:1330--1362`). The displacement for one line is

$$\begin{aligned} \delta R &= (A_R^{\rm in}\sin\theta + A_R^{\rm out}\cos\theta)\,2\sin\varphi\cos\varphi, \\ \delta N &= (A_T^{\rm in}\sin\theta + A_T^{\rm out}\cos\theta)\,\cos 2\varphi, \\ \delta E &= (A_T^{\rm in}\cos\theta + A_T^{\rm out}\sin\theta)\,\sin\varphi, \end{aligned} \tag{10}$$

(`GnssCorrections.cpp:1366--1373`), summed over the table and rotated radial/north/east → ECEF. The reference amplitudes (mm) include K_1 : (12.00, −0.80, −0.67, −0.03) and P_1 : (−1.23, −0.07, 0.06, 0.01) in the order $(A_R^{\rm in}, A_R^{\rm out}, A_T^{\rm in}, A_T^{\rm out})$ (`GnssCorrections.cpp:1346, 1350`).

Long-period band (Table 7.3b). Five zonal lines, with no longitude or τ dependence and no east component (`GnssCorrections.cpp:1384--1389`). The radial weight is the zonal Legendre form $(3\sin^2\varphi - 1)/2$ and the north weight is $\sin 2\varphi$:

$$\delta R = (A_R^{\rm in}\cos\theta + A_R^{\rm out}\sin\theta)\,\frac{3\sin^2\varphi - 1}{2}, \qquad \delta N = (A_T^{\rm in}\cos\theta + A_T^{\rm out}\sin\theta)\,\sin 2\varphi, \tag{11}$$

(`GnssCorrections.cpp:1393, 1399--1400`). The full solid-tide displacement is $\boldsymbol{d}_{\rm SET} = \boldsymbol{d}^{(1)}_{\rm SET}$ + ST1IDIU + ST1ISEM + ST1L1 + STEP2DIU + STEP2LON (`GnssCorrections.cpp:1448--1463`).

1.3 Ocean tidal loading (OTL)

Ocean tidal loading is the elastic response of the crust to the redistributed mass of the ocean tides; it is significant (∼cm) near coasts and is synthesised as a harmonic sum over the eleven principal constituents M2, S2, N2, K2, K1, O1, P1, Q1, Mf, Mm, Ssa (`OceanTideModel.hpp`, `OTL_CONSTITUENT_NAMES`). Each constituent k contributes, on each local component $c \in \{\text{up}, \text{west}, \text{south}\}$,

$$\boxed{dr_c(t) \;=\; \sum_{k=1}^{11} A_{c,k}\, \cos\!\big(\chi_k(t) - \phi_{c,k}\big),} \tag{12}$$

where $A_{c,k}$ is the station amplitude, $\phi_{c,k}$ the Greenwich phase lag, and $\chi_k(t)$ the astronomical argument (`OceanTideModel.cpp:224--233`). The internal accumulation is in the (up, west, south) frame; the conversion to ENU is $(E, N, U) = (-\text{west}, -\text{south}, \text{up})$ (`OceanTideModel.cpp:235--236`), then ENU→ECEF (`OceanTideModel.cpp:240`).

1.3.1 Astronomical argument and the Doodson–Warburg phase bias

The argument χ_k is assembled from Doodson multipliers $[\,\tau, s, h, p, N', p_s\,]$ acting on the variables $[\,\text{GMST} + \pi,\ s,\ h,\ p,\ N',\ p_s\,]$, themselves derived from the Delaunay fundamental arguments $s = F + \Omega,\ h = s - D,\ p = s - l,\ N' = -\Omega$ (`OceanTideModel.cpp:142--176`). Critically, the canonical tide argument to which both the BLQ Greenwich phase lags and the EOT20/FES complex coefficients are referenced carries a *species-dependent Doodson–Warburg phase bias*: a quarter cycle ($\pm 90°$) for the diurnal band, because the Cartwright–Tayler–Edden tide-generating potential refers to a *sine* rather than a cosine when (degree + n_τ) is odd. The intended bias column (cycles) is

M2	S2	N2	K2	K1	O1	P1	Q1	Mf	Mm	Ssa
0	0	0	0	+0.25	−0.25	−0.25	−0.25	0	0	0

(13)

applied as $\chi_k \mathrel{+}= 2\pi\,(\text{bias}_k)$ (`OceanTideModel.cpp:178--183`; `OceanTideModel.hpp`, `CONSTITUENT_PHASE_BIAS_CYCLES`). **Omitting this bias phases every diurnal constituent (K1/O1/P1/Q1) by a full** 90°, mismodelling the entire diurnal OTL contribution — this is the intended oracle resolution of the OTL-diurnal-phase question. The bias *must* be present and carry the species signs of Equation (13).

1.3.2 Coefficient sources: BLQ and the EOT20 grid

Per-station amplitudes/phases come either from a station BLQ block or, when none is supplied, are synthesised by bilinear interpolation of the global EOT20 complex load grid at the receiver's a priori position (`ModelGeometry.cpp:99--105`). The EOT20 netCDF packs the real and imaginary load coefficients as `int16` with a `scale_factor`/`add_offset`; the netCDF C library does *not* auto-apply these, so they are unpacked manually as $v = v_{\text{int16}} \cdot \text{scale} + \text{offset}$ (the `scale`$\approx 5.34 \times 10^{-5}$ m/LSB; omitting it inflates amplitudes by ∼20000×) (`OceanTideModel.cpp:443--461, 528--529`). The complex load $Z = \text{Re} + i\,\text{Im}$ yields the real signal $h(t) = \text{Re}\{Z\, e^{i\chi}\} = A\cos(\chi - \phi)$ with

$$A = \sqrt{\text{Re}^2 + \text{Im}^2}, \qquad \phi = \text{atan2}(-\text{Im}, \text{Re}) \tag{14}$$

(ϕ in BLQ format being the Greenwich phase lag in degrees, `OceanTideModel.cpp:551--556`). Because EOT20 orders its 17 constituents differently from the BLQ convention, the synthesis remaps the BLQ slot k to the EOT20 index via $\{4, 13, 8, 3, 2, 9, 10, 11, 6, 7, 15\}$ (`OceanTideModel.cpp:541`).

1.4 Pole tide (rotational deformation, IERS 7.1.4)

The (solid) pole tide is the elastic deformation driven by the centrifugal potential of polar motion: as the instantaneous rotation pole wanders relative to its secular mean, the bulge it raises moves with it. The wobble is the polar-motion offset from the IERS 2018 secular mean pole. In milliarcseconds, with t_y the decimal year,

$$\bar{x}_p = 55.0 + 1.677\,(t_y - 2000), \qquad \bar{y}_p = 320.5 + 3.460\,(t_y - 2000) \tag{15}$$

(`GnssCorrections.cpp:935--936`); this linear model is the convention to which the post-2018 IGS products consumed by the engine are referenced. The wobble components (arcseconds) are

$$m_1 = x_p - \bar{x}_p, \qquad m_2 = -\big(y_p - \bar{y}_p\big), \tag{16}$$

with x_p, y_p the polar-motion values linearly interpolated from the ERP table at the query epoch (`GnssCorrections.cpp:949--950`; the $\bar{x}_p, \bar{y}_p$ millarcsecond means are scaled to arcseconds in the difference). **The wobble must be carried in arcseconds, not radians**: the displacement coefficients below are metre-valued per arcsecond, so forming m_1, m_2 in radians underscales the whole pole tide by $\sim 4.85 \times 10^{-6}$ (a previously-latent bug noted at `GnssCorrections.cpp:943--948`).

IERS 2010 Eq. 7.24 gives the rotational-deformation displacement in colatitude $\theta = 90^\circ - \varphi$; substituting geodetic latitude ($\sin 2\theta = \sin 2\varphi$, $\cos 2\theta = -\cos 2\varphi$, $\cos\theta = \sin\varphi$, north $= -S_\theta$) yields the radial, north, and east components (metres):

$$\boxed{\begin{aligned} \boldsymbol{d}^R_{\mathrm{PT}} &= -0.033\,\sin 2\varphi\,\big(m_1\cos\lambda + m_2\sin\lambda\big), \\ \boldsymbol{d}^N_{\mathrm{PT}} &= -0.009\,\cos 2\varphi\,\big(m_1\cos\lambda + m_2\sin\lambda\big), \\ \boldsymbol{d}^E_{\mathrm{PT}} &= +0.009\,\sin\varphi\,\big(m_1\sin\lambda - m_2\cos\lambda\big), \end{aligned}} \tag{17}$$

(`GnssCorrections.cpp:967--969`). **The east factor is** $\sin\varphi$ **(**$= \cos\theta$**), not** $\cos\varphi$: this is the intended oracle resolution of the pole-tide east-term question. An east factor of $\cos\varphi$ is internally inconsistent with the geodetic-φ radial/north terms ($\sim$1.48$\times$ scale error at mid-latitude, and the wrong latitude dependence — vanishing at the equator instead of the pole). The local (R, N, E) vector is rotated into ECEF at `GnssCorrections.cpp:986--988`.

1.5 Ocean pole tide (Desai 2002, IERS 7.1.5)

Polar motion also redistributes ocean mass (the ocean pole tide), whose load deforms the crust by a further sub-millimetre to $\sim$2 mm. It adds onto Equation (17) when the station's six Desai grid load coefficients are available (`GnssCorrections.cpp:971--984`). Following Desai (2002) [1], the same wobble m_1, m_2 drives a complex pole-tide coefficient

$$c^R = m_1\,\gamma_2^R + m_2\,\gamma_2^I, \tag{18}$$

$$c^I = m_2\,\gamma_2^R - m_1\,\gamma_2^I, \tag{19}$$

with the IERS/Desai load Love combination $\gamma_2^R = 0.6870$, $\gamma_2^I = 0.0036$ (`GnssCorrections.cpp:822--823, 843--844`). Each of the six per-station coefficients u_c^R, u_c^I (north, east, radial real and imaginary parts), bilinearly interpolated from the Desai grid, gives the local displacement

$$\boxed{\boldsymbol{d}^c_{\mathrm{OPT}} = \kappa\,\big(c^R\,u_c^R + c^I\,u_c^I\big), \qquad c \in \{E, N, U\},} \tag{20}$$

(`GnssCorrections.cpp:845--847`). The scale κ folds in the IERS load-deformation constants and the arcsec→rad conversion (so that the wobble may stay in arcseconds, consistent with Equation (16)):

$$\kappa = \frac{4\pi\, G\, a\, \rho_w\, H_p}{3\, g_E}\, \frac{\pi}{648000}, \qquad H_p = \sqrt{\tfrac{8\pi}{15}}\, \frac{\Omega^2 a^4}{GM_\oplus}, \tag{21}$$

with $\rho_w = 1025\ \mathrm{kg\,m^{-3}}$, $a = 6378136.6$ m, $g_E = 9.7803278\ \mathrm{m\,s^{-2}}$, $\Omega = 7.292115 \times 10^{-5}\ \mathrm{rad\,s^{-1}}$, $G = 6.67428 \times 10^{-11}$, $GM_\oplus = 3.986004418 \times 10^{14}$ (`GnssCorrections.cpp:816--836`). The whole ocean-pole-tide path reproduces the NGS M-PAGES implementation and the independent Python oracle `scripts/ocean_pole_tide_reference.py` to $\sim 10^{-12}$ (`GnssCorrections.cpp:808--813`).

1.6 Atmospheric pressure loading — a known gap

Honesty note, in the spirit of the SSR document's flag of the iono-not-SSR gap. Atmospheric pressure loading (APL) — the crustal response to surface-pressure redistribution, together with the S1/S2 atmospheric thermal tides — is the *largest unmodelled* site displacement here, $\sim$2–20 mm predominantly vertical. It is presently **absent**. The function `computeAtmosphericLoadingCorrection` (`GnssCorrections.cpp:1142--1152`) is a 1 mm cosine stub (amplitude 0.001, a bare diurnal-times-$\cos\varphi$ placeholder) and is *uncalled*. Moreover the struct field named `atm_loading_corr` does *not* carry an atmospheric term at all: it carries the OTL projection $\boldsymbol{d}_{\mathrm{OTL}} \cdot (-\boldsymbol{e})$ (`ModelGeometry.cpp:825`). A genuine APL model (e.g. a pressure-driven Green's-function or gridded loading product, plus the S1/S2 tidal harmonics) is the leading candidate for the next site-displacement term; until it lands, $\boldsymbol{d}_{\mathrm{APL}} = \mathbf{0}$ in Equation (1).

1.7 Integration into the measurement model

Operationally, the per-epoch geometry builder evaluates each enabled displacement, projects it via Equation (2), and the modelled-range assembly adds the scalar corrections. In `ModelGeometry.cpp` the OTL ECEF displacement is formed (`:822--824`) and projected to `atm_loading_corr` (`:825`); the solid tide is evaluated against the cached Sun/Moon positions and already returns its LOS-projected scalar (`:827--831`). In `PppModel.cpp` the pole tide (and, when enabled, the ocean pole tide additively inside it) is projected by `computePoleTideProjection` (`:1402--1408`) and the base modelled value sums `+ atm_loading_corr + solid_tide` (`:1455--1456`) while the pair processors add `+ pole_tide_corr` (`:1510, 1747`). The resulting corrected geometric range is precisely the $\rho(\boldsymbol{r})$ that enters the linearised range once $\boldsymbol{r}$ is replaced by Equation (1). Because each term is gated by its own config flag and contributes a zero vector when off, the differenced observables of the BSSD and double-difference formulations inherit whatever residual of these corrections survives differencing: solid Earth tide and pole tide are highly spatially correlated and largely cancel over short baselines, whereas ocean loading and the (absent) atmospheric loading do not cancel and remain the dominant differential site-displacement error on long baselines.

References

[1] Shailen D. Desai. "Observing the pole tide with satellite altimetry". In: *Journal of Geophysical Research: Oceans* 107.C11 (2002), p. 3186. DOI: 10.1029/2001JC001224.

[2] Gérard Petit and Brian Luzum. *IERS Conventions (2010)*. IERS Technical Note 36. International Earth Rotation and Reference Systems Service (IERS). Frankfurt am Main: Verlag des Bundesamts für Kartographie und Geodäsie, 2010. ISBN: 3-89888-989-6.